

→ Inner Product (Dot Product)

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Recap

- ✓ • Matrices
- ✓ • Vectors
- ↪ • Basis and unit vectors
- ↪ • Linear combination of vectors
- Linear Transformations
- ↪ • Change of Basis →
- ↪ • Eigenvectors and Eigenvalues

eigen basis

Inner Product

Inner product (generalization of dot product) takes two vectors $\vec{u}, \vec{v}$ (from the same vector space) and provides a scalar as a result.

E.X.

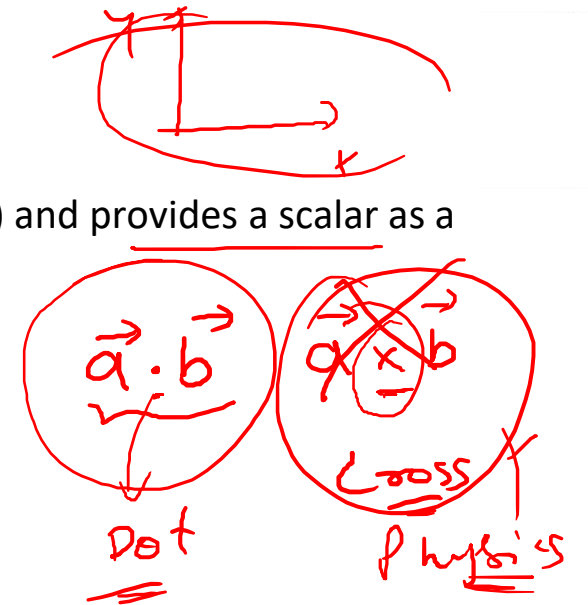
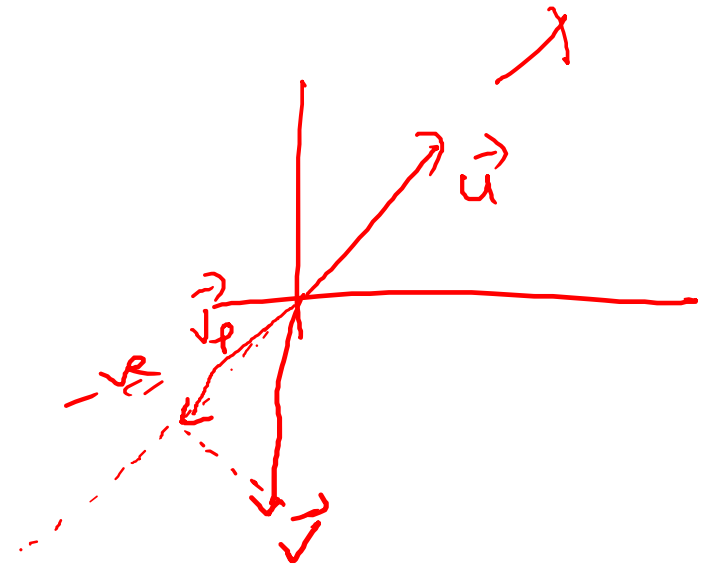
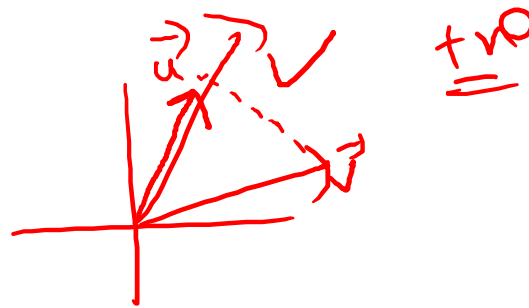
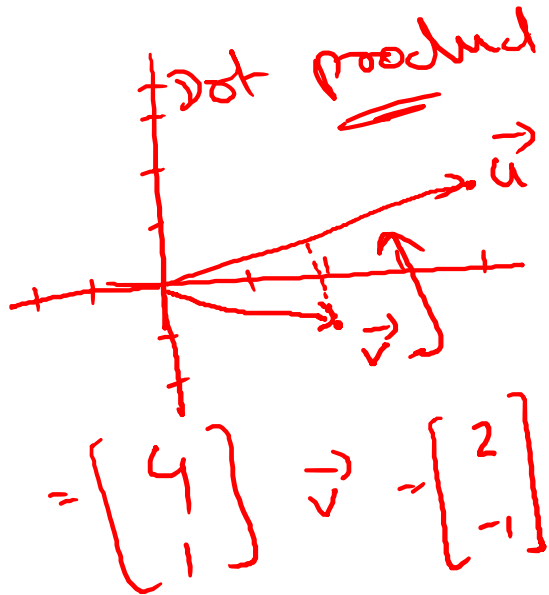
$$\vec{u} \cdot \vec{v} = \begin{bmatrix} 2 \\ 7 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 8 \\ 2 \\ 8 \end{bmatrix} = 2 \cdot 8 + 7 \cdot 2 + 1 \cdot 8 = 16 + 14 + 8 = 38$$

$$\vec{u} \cdot \vec{v} = \sum_{i=1}^n \vec{u}_i \cdot \vec{v}_i$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 1 \cdot 3 + 2 \cdot 4 = 3 + 8 = 11$$

Dot product as projections

= multiplying the projected length of $\vec{v}$ onto the $\vec{u}$



Note

- If 2 vectors are pointing in the same direction their dot product is positive.
- If 2 vectors are pointing in opposite direction their dot product is negative.
- If 2 vectors are perpendicular their dot product is zero.
- In dot product the order in which we write the vectors does not matter, hence:

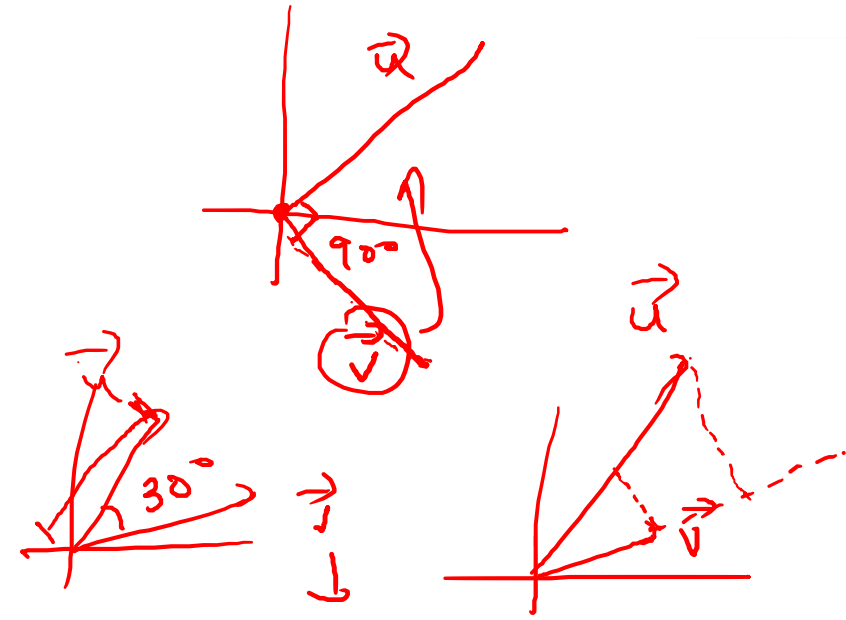
$$\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$$
Same
- The angle between two vectors can be evaluated using:

$$\vec{u} \cdot \vec{v} = \|\vec{u}\| \cdot \|\vec{v}\| \cdot \cos \theta \quad \cos \theta = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \cdot \|\vec{v}\|}$$

Where,

$\|\vec{u}\|$ is norm/length/magnitude of the vector and is calculated as: $\sqrt{\sum_{i=1}^d |\alpha_i|^2}$

It is also sometimes represented as: square root of dot product of a vector with itself: $\sqrt{\vec{u} \cdot \vec{u}}$



$$\cos \theta = \frac{\text{dot product}}{\text{norm of } \vec{u} \times \text{norm of } \vec{v}}$$

angle
between $\vec{u}$ & $\vec{v}$

$$\vec{u} = \begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix} = \begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix} \quad \sqrt{u_x^2 + u_y^2 + u_z^2} = \sqrt{\text{dot prod}}$$

$$\|\vec{u}\| = \sqrt{u_x^2 + u_y^2 + u_z^2}$$

$$\sqrt{\vec{u} \cdot \vec{u}} \Rightarrow \text{norm}$$

Dot product

Q.) You are provided with two vectors $\mathbf{u}$ and $\mathbf{v}$ such that $\mathbf{u} \cdot \mathbf{u} = 39$, $\mathbf{u} \cdot \mathbf{v} = -3$, and $\mathbf{v} \cdot \mathbf{v} = 79$. evaluate the expression:

$(\mathbf{u} + 2\mathbf{v})(3\mathbf{u} + \mathbf{v})$

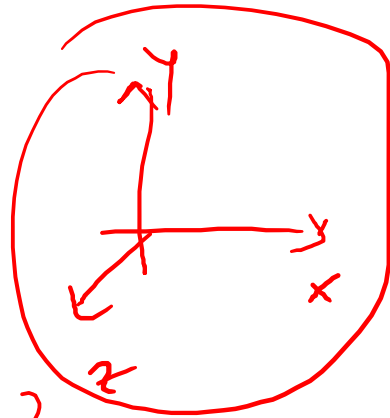
Sol.)

$$\vec{u} \cdot \vec{u} = 39$$

$$\vec{u} \cdot \vec{v} = -3$$

$$\vec{v} \cdot \vec{v} = 79$$

$$\begin{aligned} & (\vec{u} + 2\vec{v})(3\vec{u} + \vec{v}) \quad \vec{v} \cdot \vec{u} = \vec{u} \cdot \vec{v} \\ &= 3\vec{u} \cdot \vec{u} + 2\vec{v} \cdot \vec{u} + \vec{u} \cdot \vec{v} + 2\vec{v} \cdot \vec{v} \\ &= 3(39) + 3(-3) + 2(79) \\ &= \dots \end{aligned}$$



$$\vec{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad \vec{u} \cdot \vec{v} =$$

$$\vec{v} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$$

$$\cos \theta = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|}$$

Orthogonal set of Vectors

A set of vectors $\{v_1, v_2, \dots, v_k\}$ in a given vector space $\mathbb{R}^n$ is called an **orthogonal set** if all pairs of distinct vectors are orthogonal (perpendicular) that is, if:

$$v_i \cdot v_j = 0 \text{ when } i \neq j$$

E.X.

Dot product

$$\vec{u} \cdot \vec{v} = 0$$

if $\begin{cases} v_1 \cdot v_2 = 0 \\ v_2 \cdot v_3 = 0 \\ v_1 \cdot v_3 = 0 \end{cases} \rightarrow \text{Orthogonal set of vectors}$

$$v_1 = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

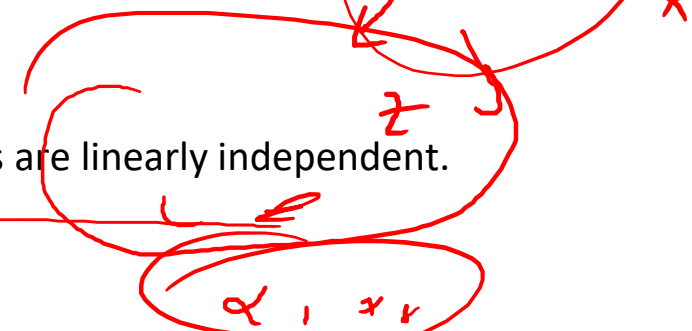
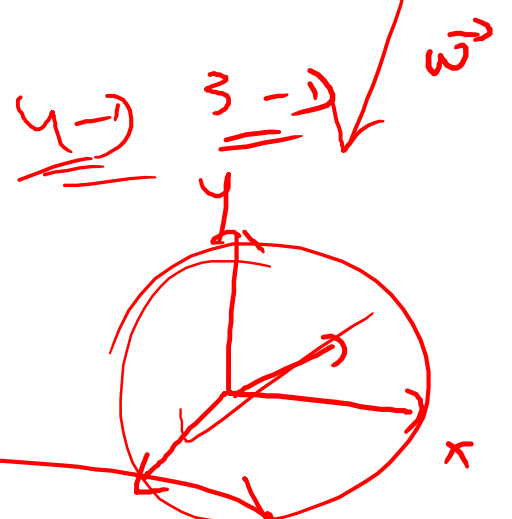
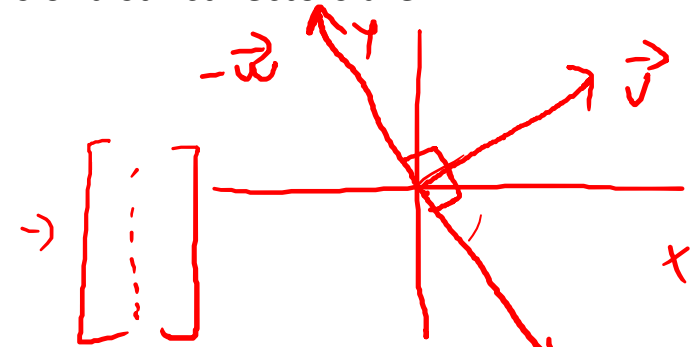
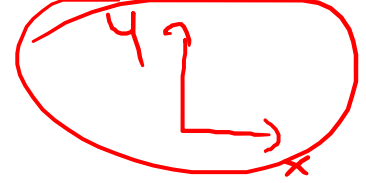
$$v_3 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$2 \cdot 0 + 1 \cdot 1 + (-1)(1) = 0$$

$$v_1 \cdot v_2 = 2 \cdot 0 + 1(1) + 1(-1) = 0$$

$$v_1 \cdot v_3 = 2 \cdot 1 + 1(-1) + (-1)(1) = 2 - 1 - 1 = 0$$

Note: If $\{v_1, v_2, \dots, v_k\}$ is an orthogonal set of non-zero vectors in $\mathbb{R}^n$, then these vectors are linearly independent.



Orthonormal set of Vectors

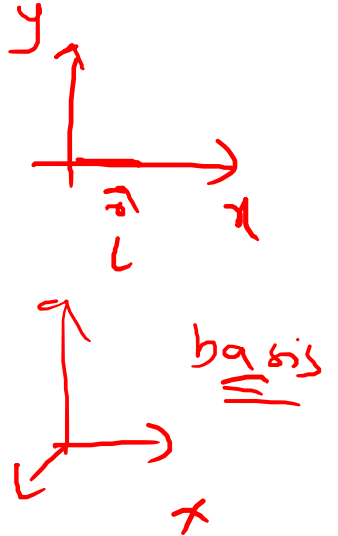
A set of vectors $\{v_1, v_2, \dots, v_k\}$ in a given vector space $\mathbb{R}^n$ is **orthonormal set** if it is an orthogonal set of **unit** vectors.

orthogonal

$$v_i \cdot v_j = \begin{cases} 0 & \text{when } i \neq j \\ 1 & \text{when } i = j \end{cases}$$

when $i \neq j$
when $i = j$

(norm) = 1



E.X.

$v \cdot v = 1$ → unit

$v_2 \cdot v_2 = 1$

$$v_1 \cdot v_2 = \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{6}} + \left(-\frac{1}{\sqrt{3}}\right) \cdot \left(\frac{2}{\sqrt{6}}\right) + \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{6}}$$

$$= \frac{1}{\sqrt{18}} - \frac{2}{\sqrt{18}} + \frac{1}{\sqrt{18}}$$

= 0

$v_1 \cdot v_2 = 0$

$$v_1 = \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

$v_2 = \begin{bmatrix} 1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}$

$v_1 \cdot v_1$

$$= \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}} + \left(-\frac{1}{\sqrt{3}}\right) \left(-\frac{1}{\sqrt{3}}\right) + \left(\frac{1}{\sqrt{3}}\right) \left(\frac{1}{\sqrt{3}}\right) = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = \frac{3}{3} = 1$$

Orthogonal Matrices

The columns of an $m \times n$ matrix Q form an orthogonal set if and only if $Q^T \cdot Q = I_n$

E.X.

$$A^T \cdot A = \text{Identity}$$

$$B = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$B^T = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$$B = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

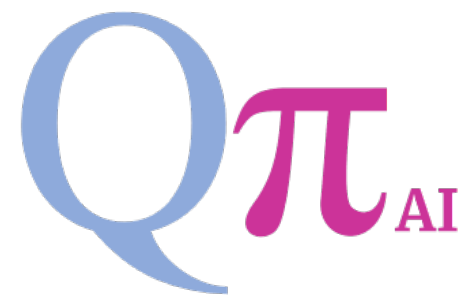
$$= \begin{bmatrix} \cos^2 \theta + \sin^2 \theta & -\cos \theta \sin \theta + \sin \theta \cos \theta \\ -\sin \theta \cos \theta + \cos \theta \sin \theta & \sin^2 \theta + \cos^2 \theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

B

$$B^T \cdot B = \text{Iden}$$

Orthogonal matrix

Don't ML



Thank you!